

GORDON GUOCHENG QIAN

Actively Seeking A **Postdoctoral** Opportunity, **2024 Spring**.

🐙 github.com/guochengqian 🌐 guochengqian.github.io ☎ (+86) 185 9209 3907 ✉ guocheng.qian@kaust.edu.sa

SUMMARY

Gordon is a Ph.D. candidate in Computer Science at KAUST under the supervision of prof. Bernard Ghanem, and is expected to graduate in Dec 2023. He has published 7 first-authored papers with over 450 citations in total. He worked at Meta Reality Lab, Microsoft Research, Megvii Research, and SenseTime Research as a research intern. His research interest lies in **Efficient 3D Vision and 3D Generation**. His representative work includes **DeepGCNs** and **PointNeXt**. He will join Snap Research as a summer intern from May to early Sep 2023, working on **3D Avatar Diffusion Models**. His long-term goal is to build AGI for digital life (virtual reality, metaverse, digital human, *etc.*).

EDUCATION

King Abdullah University of Science and Technology (KAUST)

Ph.D. student in Computer Science (3D Vision)

Jeddah, Saudi Arabia

May 2019 - Dec 2023 (Expected)

Xi'an Jiaotong University (XJTU)

Bachelor's degree in Engineering with honors, GPA: 3.9/4.0

Xi'an, China

Aug 2014 - Jul 2018

Exchange Student in Japan-Asia Youth Exchange Program

Feb 2018 - Mar 2018

Exchange Student at Hong Kong University of Science and Technology (HKUST)

Feb 2017 - May 2017

PUBLICATIONS

Full list is available at google scholar: <https://scholar.google.com/citations?user=DUDaxg4AAAAJ>

Citations: 461; H-index: 7 (Date: Mar 23, 2023)

- [1] **Guocheng Qian**, Yuchen Li, Houwen Peng, Jinjie Mai, Hasan Hammoud, Mohamed Elhoseiny, and Bernard Ghanem. **PointNeXt**: Revisiting pointnet++ with improved training and scaling strategies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022
- [2] **Guocheng Qian**, Hasan Hammoud, Guohao Li, Ali Thabet, and Bernard Ghanem. ASSANet: An anisotropic separable set abstraction for efficient point cloud representation learning. In *Advances in Neural Information Processing Systems (NeurIPS Spotlight)*, 2021
- [3] **Guocheng Qian**, Abdullellah Abualshour, Guohao Li, Ali Thabet, and Bernard Ghanem. PU-GCN: Point cloud upsampling using graph convolutional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2021
- [4] **Guocheng Qian***, G. Li*, Matthias Müller*, Itzel C. Delgadillo, Abdullellah Abualshour, Ali K. Thabet, and Bernard Ghanem. **DeepGCNs**: Making gcns go as deep as cnns. *IEEE transactions on pattern analysis and machine intelligence (T-PAMI)*, 2021
- [5] **Guocheng Qian***, G. Li*, Itzel C. Delgadillo*, Matthias Müller, Ali K. Thabet, and Bernard Ghanem. SGAS: Sequential greedy architecture search. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1617–1627, 2020
- [6] **Guocheng Qian**, Yuanhao Wang, Jinjin Gu, Chao Dong, Wolfgang Heidrich, Bernard Ghanem, and Jimmy S Ren. Rethinking learning-based demosaicing, denoising, and super-resolution pipeline. In *IEEE International Conference on Computational Photography (ICCP)*, pages 1–12. IEEE, 2022
- [7] **Guocheng Qian**, Xuanyang Zhang, Guohao Li, Chen Zhao, Yukang Chen, Xiangyu Zhang, Bernard Ghanem, and Jian Sun. When NAS meets trees: An efficient algorithm for neural architecture search. In *CVPR Workshop*, pages 2782–2787, 2022

- [8] Sicheng Chen, **Guocheng Qian**, Bernard Ghanem, Yongqing Wang, Zhou Shu, Xuefeng Zhao, Lei Yang, Xinqin Liao, and Yuanjin Zheng. Quantitative and real-time evaluation of human respiration signals with a shape-conformal wireless sensing system. *Advanced Science*, 9(32):2203460, 2022
- [9] **Guocheng Qian**, Xingdi Zhang, Abdullah Hamdi, and Bernard Ghanem. Pix4Point: Image pretrained transformers for 3d point cloud understanding. *under review*, 2023
- [10] **Guocheng Qian**, Yunyang Xiong, Haoqi Fan, Zhuang Liu, Zechun Liu, Jun Chen, Haichuan Yang, Dilin Wang, Fei Sun, Mohamed Elhoseiny, Raghuraman Krishnamoorthi, Bernard Ghanem, and Vikas Chandra. GTMNet: A global template matching convnet for video recognition. *under review*, 2023

PROFESSIONAL EXPERIENCE

Snap Research

May 2023 - Sep 2023

- Creative Vision Intern, mentored by Dr. Aliaksandr Siarohin
- Topic: “3D Avatar Diffusion Models” (ongoing)

Meta Reality Lab

Aug 2022 - Feb 2023

- AI Research Scientist Intern, mentored by Dr. Yunyang Xiong
- Paper: “GTMNet: A Global Template Matching ConvNet for Video Recognition” (under review)

Microsoft Research

Dec 2021 - May 2022

- Remote collaboration with senior researcher Dr. Houwen Peng
- Paper: “PointNeXt: Revisiting pointnet++ with improved training and scaling strategies.” (NeurIPS’22)

Megvii Technology Research

Jul 2021 - Nov 2021

- Computer Vision Intern, directed by principal researcher Dr. Xiangyu Zhang
- Paper: “When NAS Meets Trees: An Efficient Algorithm for Neural Architecture Search” (CVPRW’22)

SenseTime Research

Aug 2018 - Apr 2019

- Computer Vision Intern, mentored by senior director Dr. Jimmy S. Ren
- Paper: “Rethinking Learning-based Demosaicing, Denoising, and Super-Resolution Pipeline” (ICCP’22)

ACADEMIC EXPERIENCE

Teaching Assistant Deep Learning for Visual Computing (2020 - 2022), Design and Analysis of Algorithms (2021)

Reviewer CVPR, ECCV, ICCV, ICML, NeurIPS, AAAI, IJCAI, T-PAMI, TVCG

Organizer Tutorial on Graph Machine Learning for Computer Vision @ CVPR’22 , Workshop on Learning 3D with Multi-View Supervision @ CVPR’23

HONORS & AWARDS

KAUST CEMSE Research Excellence Award Less than a handful of students are selected.

KAUST CEMSE Dean’s List Award Awarded to only 20% of the student population by Division.

KAUST Fellowship Full tuition support, monthly living allowance, housing, and medical coverage.

National Scholarship Top 1%, first-class national scholarship in China.

Outstanding Undergraduate Highest undergraduate honor awarded to 10 selected undergraduates

Excellent Student Cadre Awarded to 3% undergraduate student with leadership

PROGRAMMING SKILLS

- **PyTorch**: PointNeXt 🌟 >400 stars, SGAS 🌟 >150 stars, DeepGCNs 🌟 >1000 stars
- **Tensorflow**: PU-GCN (🌟>100 stars)
- **Large scale GPU usage**: multi-node distributed training experience using over 100 GPUs
- **C++, CUDA**: Course projects on CUDA based image processing (🌟), and C++ based curvature estimation (🌟)
- **MATLAB**: experience in image processing, 3d reconstruction (🌟)